Computing Wasserstein Barycenter via Operator Splitting: the Method of Averaged Marginals

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Overview

I. Some context

II. Navigating the Wasserstein Barycenter Challenge

III. Our Contribution

IV. Applications

V. Conclusion

Some Context

- In applied probability, stochastic optimization, and data science, a crucial aspect is the ability to compare, summarize, and reduce the dimensionality of empirical/discrete measures

- Since these tasks rely heavily on pairwise comparisons of measures, it is essential to use an appropriate metric for accurate data analysis

- Different metrics define different barycenters of a set of measures: a barycenter is a mean element that minimizes the (weighted) sum of all its distances to the set of target measures

- When the chosen metric is the optimal transport one, and there is mass equality between the measures, the underlying barycenter is denoted by Wasserstein Barycenter (WB)

Example extracted from [1]

30 artificial images

Barycenters using

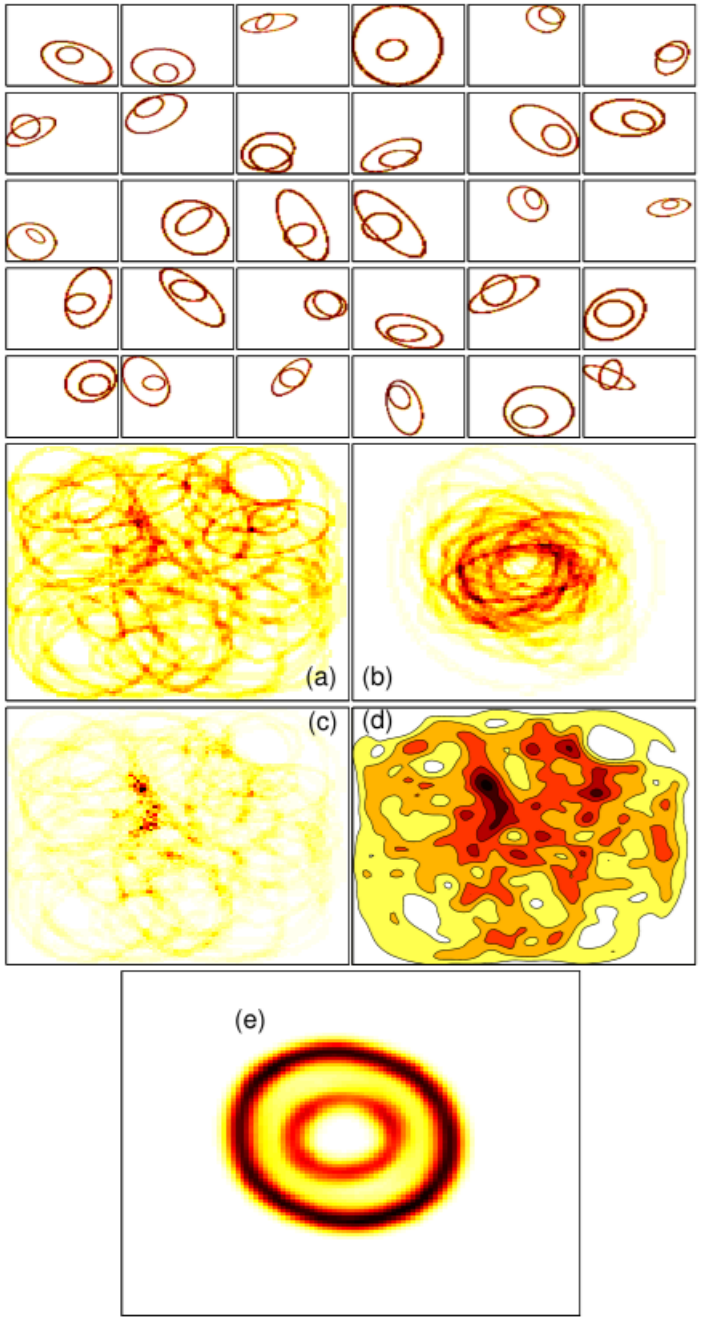
(a) Euclidean distance

(b) Euclidean + re-centering

(c) Jeffrey centroid

(d) RKHS distance

(e) 2-Wasserstein distance:
Wasserstein barycenter



[1] M. Cuturi, A. Doucet. JMLR, 2014

Navigate the Wasserstein Barycenter Challenge

The Wasserstein Distance

The Wasserstein distance - General setting

Let (Ω, P) a probabilty space and two measurable random vectors $\xi, \zeta : \Omega \mapsto \mathbb{R}^d$ such that $\mu := \xi_\# P$ and $\nu := \zeta_\# P$. Their (quadratic) 2-Wasserstein distance is

$$W_2(\xi, \zeta) := \left(\inf_{\pi \in U(\mu,\nu)} \int_{\mathbb{R}^d \times \mathbb{R}^d} \|x - y\|^2 \mathrm{d}\pi(x, y) \right)^{1/2}, \tag{WD}$$

Here $U(\mu, \nu)$ is the set of all probability measures on $\mathbb{R}^d \times \mathbb{R}^d$ having marginals μ and ν.

We focus on discrete/empirical with finite support:

$$\mu = \sum_{r=1}^{R} p_r \delta_{\xi_r} \quad \text{and} \quad \nu = \sum_{s=1}^{S} q_s \delta_{\zeta_s}$$

Wasserstein distance - discrete setting

For discrete probability measures μ and ν, their 2-Wasserstein distance W_2 is:

$$W_2(\mu, \nu) := \left(\min_{\pi \in U(\mu,\nu)} \sum_{r=1}^{R} \sum_{s=1}^{S} \|\xi_r - \zeta_s\|^2 \pi_{rs} \right)^{1/2} \tag{1}$$

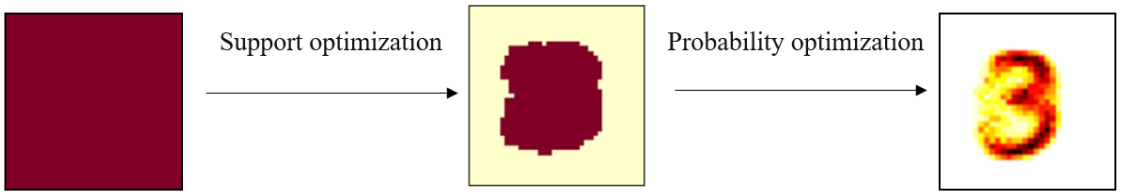
Wasserstein Barycenter (WB)

Wasserstein Barycenter problem

Given M measures $\{\nu^{(1)}, \ldots, \nu^{(M)}\} \subset \mathcal{P}(\mathbb{R}^d)$ and $\alpha \in \Delta_M$, an 2-*Wasserstein barycenter* with weights α is a solution to the following optimization problem

$$\min_{\mu \in \mathcal{P}(\mathbb{R}^d)} \sum_{m=1}^{M} \alpha_m W_2^2(\mu, \nu^{(m)}) \,. \tag{2}$$

This is a nonconvex optimization problem: the optimization is performed over the support $\texttt{supp}(\mu)$ and mass p



Huge scale Linear Program (LP)

WB problem

It boils down to solve this huge scale LP :

$$
\begin{cases}
\min\limits_{p,\pi} & \sum_{r=1}^{R}\sum_{s=1}^{S^{(1)}} c_{rs}^{(1)}\pi_{rs}^{(1)} \quad + \cdots + \quad \sum_{r=1}^{R}\sum_{s=1}^{S^{(M)}} c_{rs}^{(M)}\pi_{rs}^{(M)} \\
\text{s.t.} & \sum_{r=1}^{R}\pi_{rs}^{(1)} \qquad\qquad\qquad\qquad = q_s^{(1)}, \qquad s=1,\dots,S^{(1)} \\
& \qquad\qquad \ddots \qquad\qquad\qquad\qquad \vdots \\
& \qquad\qquad\qquad\qquad \sum_{r=1}^{R}\pi_{rs}^{(M)} = q_s^{(M)}, \qquad s=1,\dots,S^{(M)} \\
& \sum_{s=1}^{S^{(1)}}\pi_{rs}^{(1)} \qquad\qquad\qquad\qquad = p_r, \qquad r=1,\dots,R \\
& \qquad\qquad \ddots \qquad\qquad\qquad\qquad \vdots \\
& \qquad\qquad\qquad\qquad \sum_{s=1}^{S^{(M)}}\pi_{rs}^{(M)} = p_r, \qquad r=1,\dots,R \\
& p\in\Delta_R, \pi^{(1)}\geq 0 \quad \cdots \quad \pi^{(M)}\geq 0,
\end{cases}
$$

- If $M = 100$ $R = S^{(m)} = 1600$, $m = 1,\dots,M$ (corresponding to figures with 40×40 pixels), the above LP has 256 001 600 variables.
- This LP may scale exponentially in the number M of measures [2]
- Assume $S^{(m)} = S$ for all $m = 1,\dots,M$: then $R = S^M$ and the LP has $M(S)^{M+1} + (S)^M$ variables and $M(S)^M + MS$ equality constraints

[2] free support setting; S. Borgwardt. Operational Research (2022)

A vast body of the literature deals with inexact WBs

Inexact approaches

- Mostly based on reformulations via an entropic regularization: several papers by M. Cuturi, G. Peyré, G. Carlier and others
- Block-coordinate approach: fix the support and optimize the probability, then fix the probability and optimize the support [3, 4, 5]
- Other approaches [6,7,8]

Exact methods

- Methods for computing exact WBs are based on linear programming techniques [9,10]

[3] M. Cuturi, A. Doucet. JMLR, 2014
[4] J. Ye, J. Li. IEEE ICP (214)
[5] J. Ye et al. IEEE Transactions on Signal Processing (2017)
[6] G. Puccetti, L. Ruschendorf, S. Vanduffe. JMVA (2020)
[7] S. Borgwardt. Operational Research (2022)
[8] J. von Lindheim. COAP (2023)
[9] S. Borgwardt, S. Patterson (2020). INFORM J. Optimization
[10] J. Altschuler, E. Adsera. JMLR (2021)

Our Contribution

Our contribution

We provide an easy-to-implement, memory efficient and parallelizable algorithm based on the Douglas-Rachford splitting scheme to compute a solution to LPs of the form

$$\begin{cases} \min\limits_{p,\pi\geq 0} & \sum\limits_{m=1}^{M}\sum\limits_{r=1}^{R}\sum\limits_{s=1}^{S^{(m)}} d_{rs}^{(m)}\pi_{rs}^{(m)} \\ \text{s.t.} & \sum\limits_{r=1}^{R}\pi_{rs}^{(m)} = q_s^{(m)}, \quad s=1,\dots,S^{(m)},\ m=1,\dots,M \\ & \sum\limits_{s=1}^{S^{(m)}}\pi_{rs}^{(m)} = p_r, \quad r=1,\dots,R,\ m=1,\dots,M \end{cases}$$

with given $d^{(m)} \in \mathbb{R}^{R\times S^{(m)}}$

Reformulation of the LP

Separable sets

$$\Pi^{(m)} := \left\{ \pi^{(m)} \geq 0 : \sum_{r=1}^{R} \pi_{rs}^{(m)} = q_s^{(m)},\ s = 1, \ldots, S^{(m)} \right\},\ m = 1, \ldots, M \tag{3}$$

and the linear subspace

$$\mathcal{B} := \left\{ \pi = (\pi^{(1)}, \ldots, \pi^{(M)}) : \begin{array}{lll} \sum_{s=1}^{S^{(1)}} \pi_{rs}^{(1)} & = \sum_{s=1}^{S^{(2)}} \pi_{rs}^{(2)}, & \forall r \\ \sum_{s=1}^{S^{(2)}} \pi_{rs}^{(2)} & = \sum_{s=1}^{S^{(3)}} \pi_{rs}^{(3)}, & \forall r \\ & \vdots & \\ \sum_{s=1}^{S^{(M-1)}} \pi_{rs}^{(M-1)} & = \sum_{s=1}^{S^{(M)}} \pi_{rs}^{(M)}, & \forall r \end{array} \right\} \tag{4}$$

Douglas-Rachford theory

- Consider the indicator function $\mathbf{i}_C$ of a convex set C, and reformulate the problem:

$$f^{(m)}(\pi^{(m)}) := \sum_{r=1}^{R} \sum_{s=1}^{S^{(m)}} c_{rs}^{(m)} \pi_{rs}^{(m)} + \mathbf{i}_{\Pi^{(m)}}(\pi^{(m)}), \quad m = 1, \ldots, M, \tag{5}$$

- Recast problem:

$$f(\pi) := \sum_{m=1}^{M} f^{(m)}(\pi^{(m)}) \quad \text{and} \quad g(x) := \begin{cases} \mathbf{i}_{\mathcal{B}}(\pi) & \text{if balanced} \\ \gamma\, \mathtt{dist}_{\mathcal{B}}(\pi) & \text{if unbalanced.} \end{cases} \tag{6}$$

- Since f is polyhedral, it follows that computing one of its solutions is equivalent to

$$\text{find} \quad \pi \quad \text{such that} \quad 0 \in \partial f(\pi) + \partial g(\pi). \tag{7}$$

Douglas-Rachford algorithm

Given initial point $\theta^0 = (\theta^{(1),0}, \ldots, \theta^{(M),0})$ and prox-parameter $\rho > 0$:

$$\begin{cases} \pi^{k+1} & = & \mathrm{prox}_{g/\rho}(\theta^k) \\ \hat{\pi}^{k+1} & = & \mathrm{prox}_{f/\rho}(2\pi^{k+1} - \theta^k) \\ \theta^{k+1} & = & \theta^k + \hat{\pi}^{k+1} - \pi^{k+1} \end{cases} \tag{8}$$

The Method of Averaged Marginals (MAM)

The Douglas Rachford and The MAM Algorithm

$$\begin{cases} \pi^{k+1} & = & \texttt{Proj}_{\mathcal{B}}(\theta^k) \xrightarrow{\text{explicit}} \pi_{rs}^{(m)} := \theta_{rs}^{(m)} + \frac{(p_r^k - p_r^{(m)})}{S^{(m)}} \\ \hat{\pi}^{k+1} & := & \begin{pmatrix} \hat{\pi}_{1s}^{(m)} \\ \vdots \\ \hat{\pi}_{Rs}^{(m)} \end{pmatrix} = \texttt{Proj}_{\Delta_R(q_s^{(m)})} \begin{pmatrix} y_{1s} - \frac{1}{\rho} c_{1s}^{(m)} \\ \vdots \\ y_{Rs} - \frac{1}{\rho} c_{Rs}^{(m)} \end{pmatrix}, \quad s = 1, \ldots, S^{(m)} \\ \theta^{k+1} & = & \theta^k + \hat{\pi}^{k+1} - \pi^{k+1} \end{cases}$$

- At every iteration the barycenter approximation p^k is a weighted average of the M marginals $p^{(m)} := \sum_{s=1}^{S^{(m)}} \theta_{rs}^{(m)}$.
- The projection onto the simplex can be accomplished in parallel, exactly and efficiently by specialized algorithms[11].
- The whole sequence converges to an exact barycenter.

[11] see L. Condat 2016; computation in $O(R \log(R))$.

Unbalanced Wasserstein barycenters

Linear subspace of balanced plans:

$$\mathcal{B} = \left\{ \pi : \sum_{s=1}^{S^1} \pi^1_{rs} = \cdots = \sum_{s=1}^{S^M} \pi^M_{rs}, \quad r = 1, \ldots, R \right\}$$

ν^m, $m = 1, \ldots, M$, have equal masses

Balanced WB

$$\begin{cases} \min\limits_{\pi \in \mathcal{B}} & \sum\limits_{m=1}^{M} \langle d^m, \pi^m \rangle \\ \text{s.t.} & \pi^1 \in \Pi^m \\ & \vdots \\ & \pi^M \in \Pi^M \end{cases}$$

ν^m, $m = 1, \ldots, M$, have different masses

Unbalanced WB ($\gamma > 0$)

$$\begin{cases} \min\limits_{\pi} & \sum\limits_{m=1}^{M} \langle d^m, \pi^m \rangle + \gamma \, \mathtt{dist}_{\mathcal{B}}(\pi) \\ \text{s.t.} & \pi^1 \in \Pi^m \\ & \vdots \\ & \pi^M \in \Pi^M \end{cases}$$

MAM can be easily adapted to deal with both balanced or unbalanced WBs[12]

[12] the new proximal is known to be solved analytically; Bauschke and P. L. Combettes, Spinger 2017

Applications

Qualitative comparisons - Fixed support $R = 1\,600$

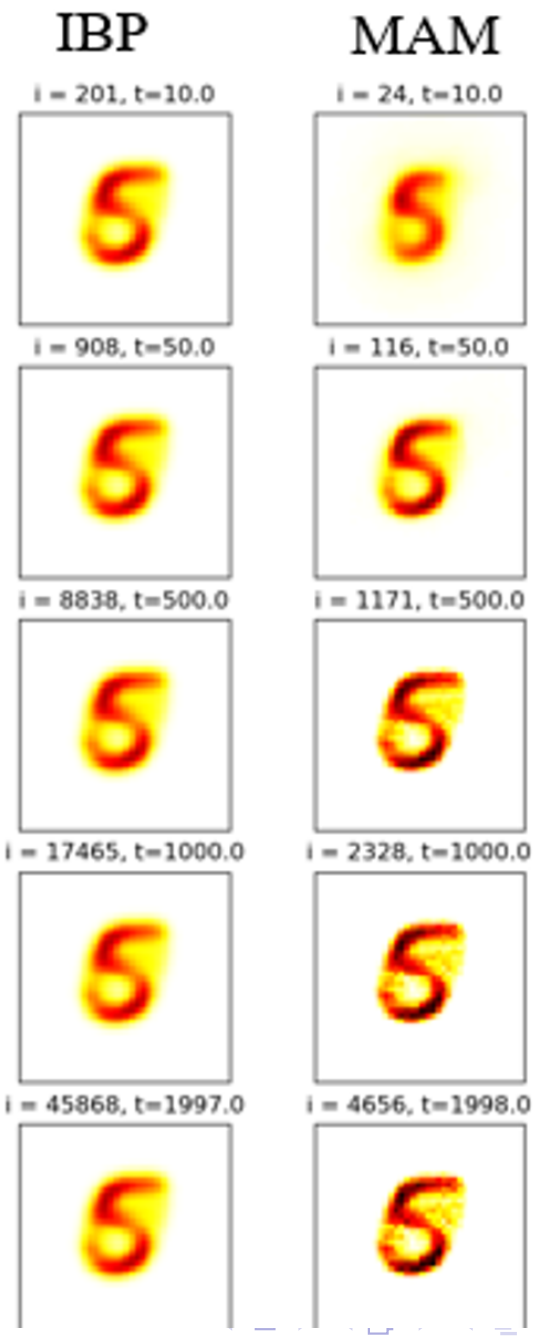
We benchmark MAM, randomized MAM, and IBP (Iterative Bregman Projection of [13]) on the MNIST database with $M = 100$ images of 40×40 pixels. LP's dimension: 256 001 600 variables and 320 000 constraints

[13] [J.-D. Benamou et al. SIAM Journal on Scientific Computing. (2015)]

Qualitative comparisons - Fixed support $R = 1\,600$

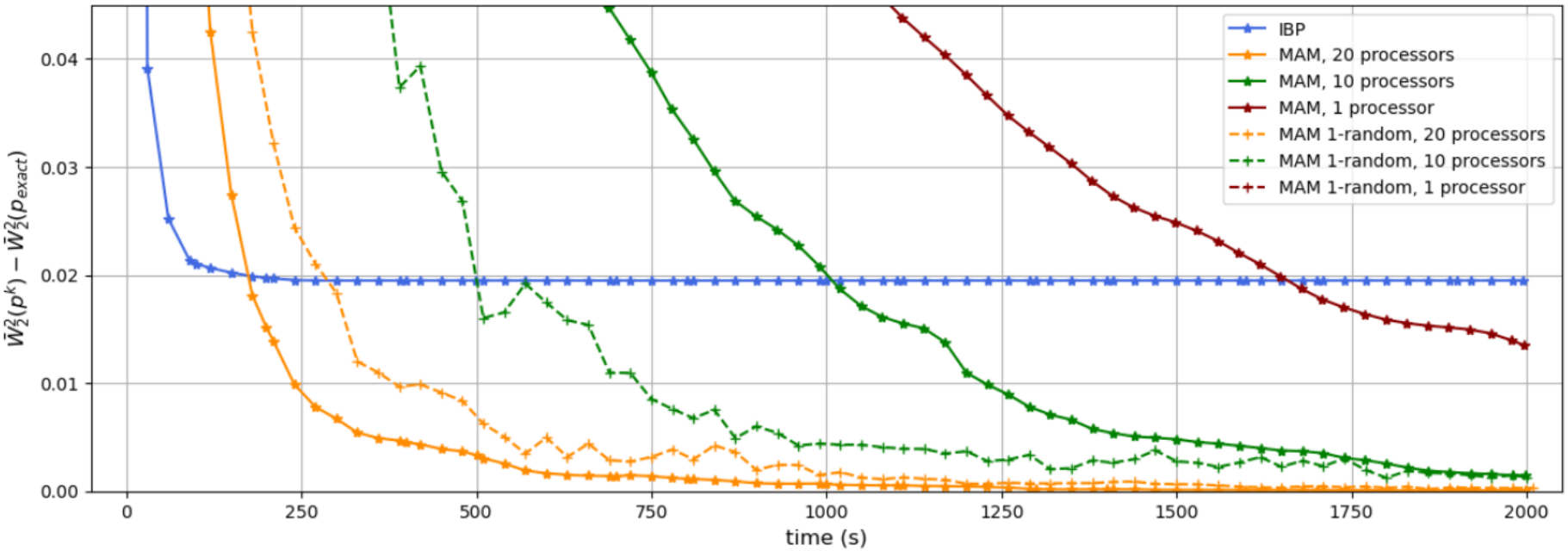
WB

Qualitative comparisons - Fixed support $R = 1\,600$



WB

Quantitative comparisons - Fixed support $R = 1\,600$



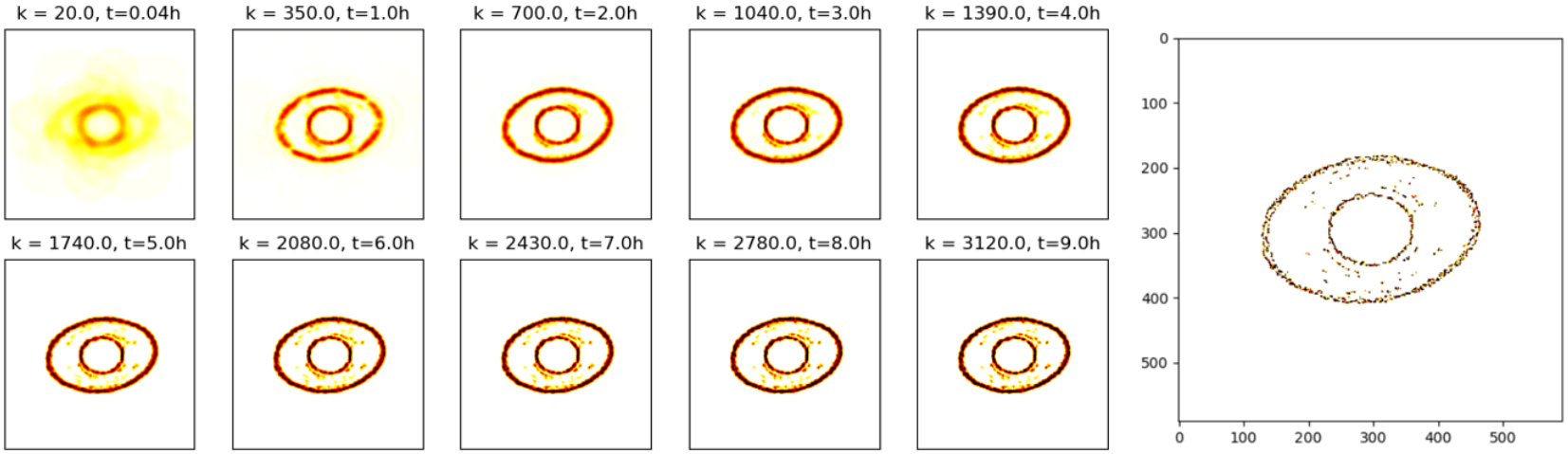
Evolution with respect to time of the difference between the Wasserstein barycenter distance of an approximation, $\bar{W}_2^2(p^k)$, and the Wasserstein barycentric distance of the exact solution $\bar{W}_2^2(p_{exact})$ given by the LP. The time step between two points is 30 seconds

Exact free-support resolution

The dataset we use is the one from [14]: $M = 10$ images of 60×60 pixels
LP's dimension: $1.2574 \cdot 10^{10}$ variables and $3.5288 \cdot 10^{6}$ constraints
We compare with the dedicated solver of Altschuler and Boix-Adsera, available at [15]

- All measures share the same finite support: suppose that all measures $\nu^{(m)}$ are supported on a d-dimensional regular grid of integer step sizes in each direction, each coordinate going from 1 to K: $S^{(m)} = S = K^d$.

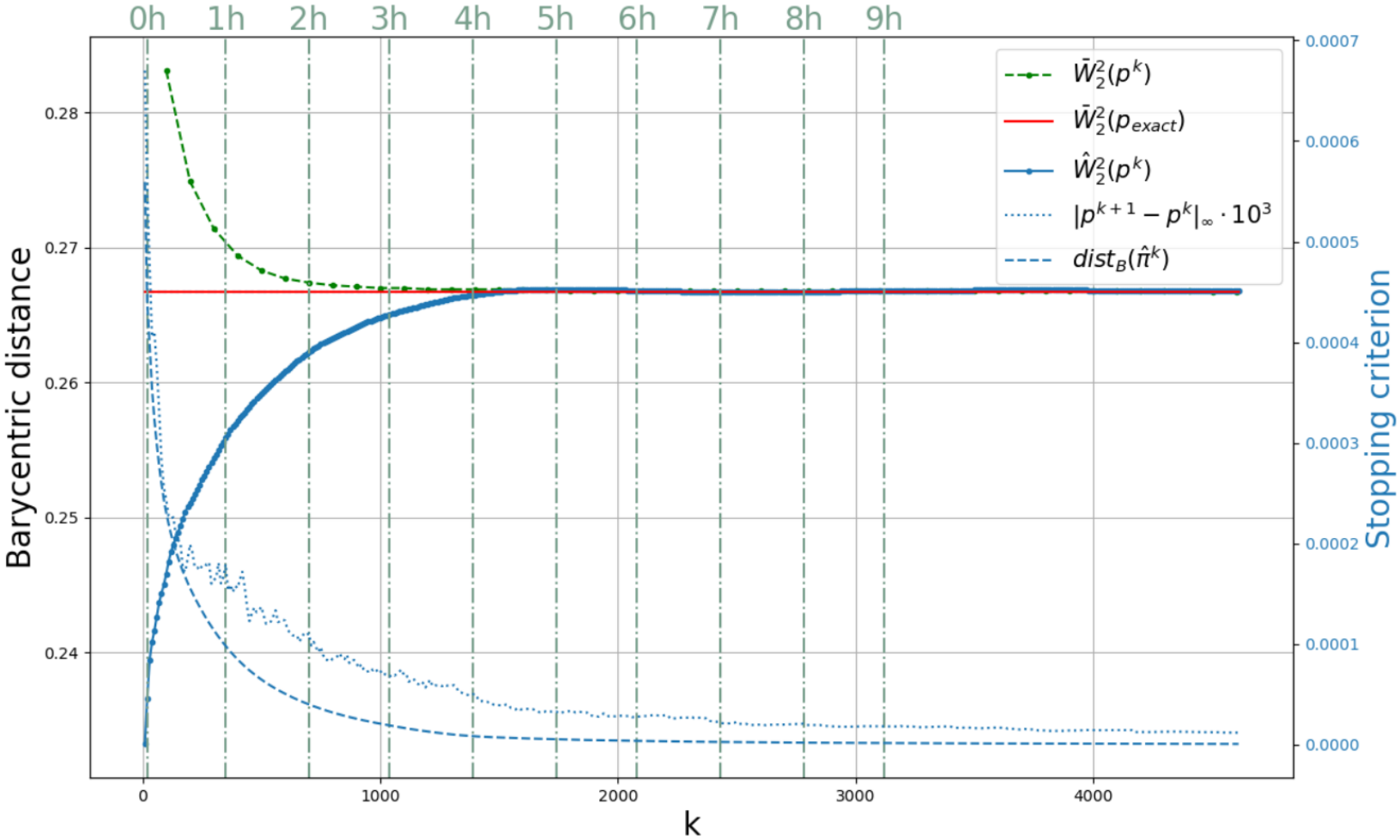
- The measures are evenly weighted $\alpha_m = \frac{1}{M}$, $m = 1, \ldots, M$



Evolution of the approximated MAM barycenter with time in regards with the exact barycenter of the Altschuler and Bois-Adsera algorithm computed in 3.5 hours [16]

MAM can solve larger problems than the method Altschuler and Boix-Adsera

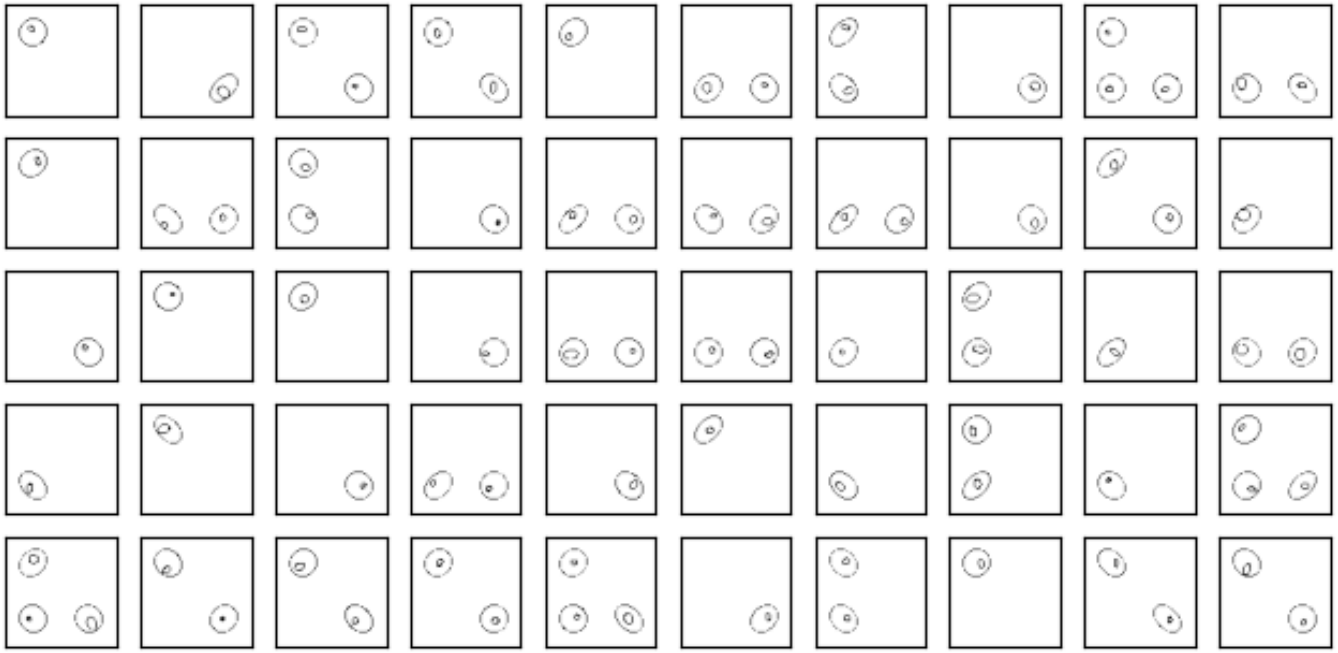
[14] J. M. Altschuler and E. Boix-Adsera. JMLR (2021)
[15] `https://github.com/eboix/high_precision_barycenters`
[16] S. Borgwardt, S. Patterson (2020). INFORM J. Optimization



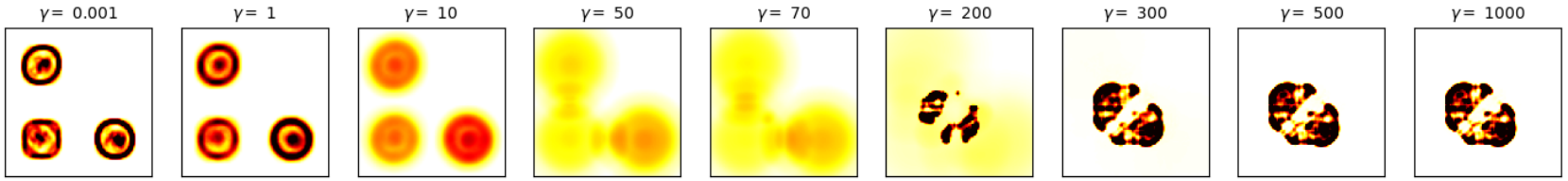
The optimal value of the WB problem is 0.2666

After 1 hour of processing, MAM had a barycenter distance of 0.2702, which improved to 0.2667 after 3.5 hours, when the solver of Altschuler and Boix-Adsera halts

Unbalanced WB



$$\begin{cases} \min\limits_{\pi} & \sum_{m=1}^{M} \langle d^{(m)}, \pi^{(m)} \rangle + \gamma \mathtt{dist}_{\mathcal{B}}(\pi) \\ \text{s.t.} & \pi^{(1)} \in \Pi^{(m)}, \ldots, \pi^{(M)} \in \Pi^{(M)} \end{cases}$$



Constrained Wasserstein barycenters

Suppose the probability vector p is constrained to a closed convex set $X \subset \mathbb{R}^R$:

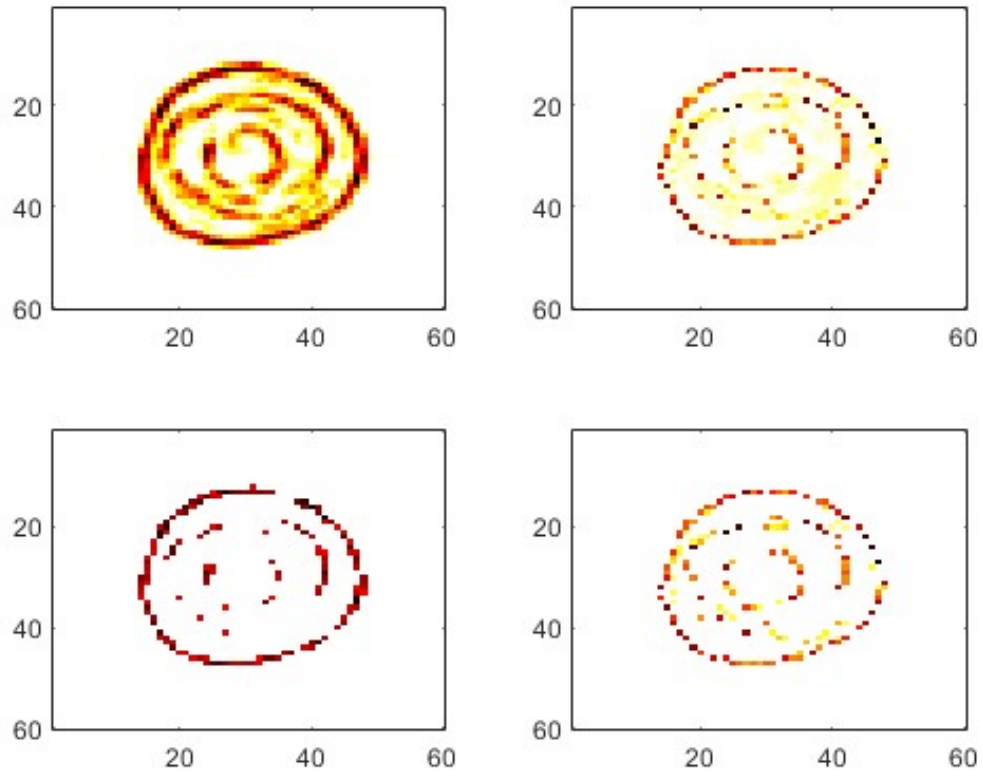
$$\begin{cases} \min\limits_{p,\pi \geq 0} & \sum\limits_{m=1}^{M} \langle d^m, \pi^m \rangle \\ \text{s.t.} & \sum\limits_{r=1}^{R} \pi_{rs}^m = q_s^m, \quad s = 1, \ldots, S^m, \; m = 1, \ldots, M \\ & \sum\limits_{s=1}^{S^m} \pi_{rs}^m = p_r, \quad r = 1, \ldots, R, \; m = 1, \ldots, M \\ & p \in X \end{cases}$$

We can easily adapt MAM to compute constrained WBS

What if X is not convex ?

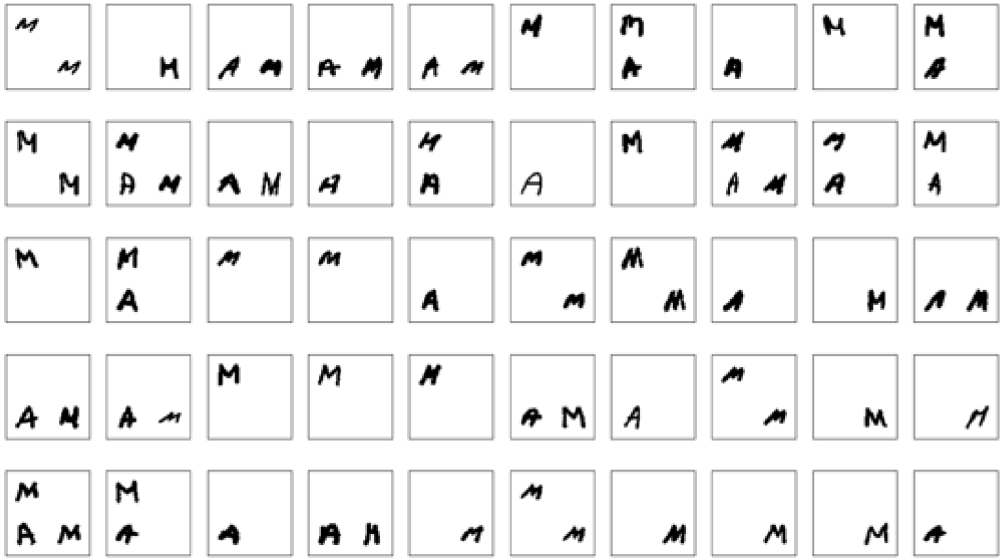
Constrained Wasserstein barycenters

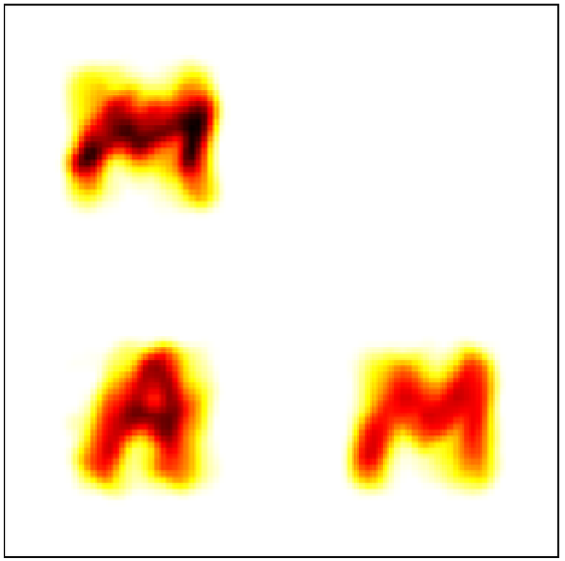
Extension to the unconvex case with $X_k = \{p \in \mathbb{R}^R : \|p\|_0 = k\}$ for a triple nested dataset (25 images):



The aim of this application is to extract a sharper barycenter
Top left: MAM balanced barycenter
Top right: MAM with the unconvex constrain

Bottom: computed barycenters after a direct projection onto X. The loss mass is 80% for the classical MAM and only 15% for the constrained MAM.





Take-away messages

- New easy-to-implement and memory efficient algorithm for computing WBs, which is parallelizable and can run in a randomized manner if necessary
- It can be applied to both balanced WB and unbalanced WB problems upon setting a single parameter
- It can be applied to the free or fixed-support settings
- The method is submitted (after review) in SIAM Journal on Mathematics and Data Science `https://arxiv.org/pdf/2309.05315.pdf`
- Our Python code is freely available at `https://ifpen-gitlab.appcollaboratif.fr/detocs/mam_wb`

Thank you!

D. Mimouni, P. Malisani, J. Zhu, W. de Oliveira. Computing Wasserstein barycenter via operator splitting: the method of averaged marginals, https://arxiv.org/pdf/2309.05315.pdf, 2023

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Annexes

Special setting: grid-structured data

- All measures share the same finite support: suppose that all measures $\nu^{(m)}$ are supported on a d-dimensional regular grid of integer step sizes in each direction, each coordinate going from 1 to K: $S^{(m)} = S = K^d$, and $\texttt{supp}(\nu^{(m)}) := \{\zeta_1, \ldots, \zeta_S\}$, $m = 1, \ldots, M$
- The measures are evenly weighted $\alpha_m = \frac{1}{M}$, $m = 1, \ldots, M$
- Then $\texttt{supp}(\mu)$ has at most

$$R \leq ((K-1)M + 1)^d$$

points, as the finer grid only runs between the boundary points [17]

This significantly reduces the LP's dimension

[17] S. Borgwardt, S. Patterson (2020). INFORM J. Optimization

Example (LP's dimensions)

Consider the case: $M = 10$, $d = 2$, $K = 40 \Rightarrow S = 1600$

data	$\lvert\mathtt{supp}(\mu)\rvert$ R	# variables $(MS+1)R$	# eq. constraints $(S+R)M$
general	$1.0995 \cdot 10^{32}$	$1.7593 \cdot 10^{36}$	$1.0995 \cdot 10^{33}$
grid-structured	152881	$2.4462 \cdot 10^{9}$	1544810

- In contrast to the worst-case, exponentially sized possible support set, there always exists a WB $\bar{\mu}$ with provably sparse support

$$|\mathtt{supp}(\bar{\mu})| \leq \sum_{m=1}^{M} S^{(m)} - M + 1$$

- For the above example $|\mathtt{supp}(\bar{\mu})| \leq 15991$

- This fact motivates heuristics for computing inexact WBs: fixed-support approaches, which generally fix R to $\sum_{m=1}^{M} S^{(m)} - M + 1$ (or fewer) points

The Method of Averaged Marginals - MAM

Unbalanced Wasserstein barycenter

Algorithm

Input: initial plan $\pi = (\pi^1, \dots, \pi^m)$ and parameters $\rho, \gamma > 0$
Define $a_m \leftarrow (\frac{1}{S^m})/(\sum_{j=1}^{M} \frac{1}{S^j})$ and set $p^m \leftarrow \sum_{s=1}^{S^m} \pi^m_{rs}$, $m = 1, \dots, M$

while not converged **do**

  $p \leftarrow \sum_{m=1}^{M} a_m p^m$ ▷ Average the marginals

  Set $t \leftarrow 1$ if $\rho \sqrt{\sum_{m=1}^{M} \frac{\|p - p^m\|^2}{S^m}} \leq \gamma$; else $t \leftarrow \gamma / \left(\rho \sqrt{\sum_{m=1}^{M} \frac{\|p - p^m\|^2}{S^m}} \right)$

  for $m = 1, \dots, M$ **do**
    for $s = 1, \dots, S^m$ **do**
      $\pi^m_{:s} \leftarrow \mathtt{Proj}_{\Delta(q^m_s)} \left(\pi^m_{:s} + 2t \frac{p - p^m}{S^m} - \frac{1}{\rho} d^m_{:s} \right) - t \frac{p - p^m}{S^m}$
    end for
    $p^m \leftarrow \sum_{s=1}^{S^m} \theta^m_{rs}$ ▷ Update the mth marginal
  end for

end while

Set $\gamma = \infty$ to compute balanced WB (if the measures are balanced)
Otherwise, choose $\gamma \in (0, \infty)$ to compute unbalanced WB

The Method of Averaged Marginals - MAM

Constrained setting

Algorithm

Input: initial plan $\pi = (\pi^1, \dots, \pi^m)$ and parameter $\rho > 0$

Define $a_m \leftarrow (\frac{1}{S^m})/(\sum_{j=1}^{M} \frac{1}{S^j})$ and set $p^m \leftarrow \sum_{s=1}^{S^m} \pi^m_{rs}$, $m = 1, \dots, M$

while not converged **do**

  $p \leftarrow \texttt{Proj}_X \left(\sum_{m=1}^{M} a_m p^m \right)$ ▷ Average the marginals

  for $m = 1, \dots, M$ **do**

    for $s = 1, \dots, S^m$ **do**

      $\pi^m_{:s} \leftarrow \texttt{Proj}_{\Delta(q^m_s)} \left(\pi^m_{:s} + 2\frac{p - p^m}{S^m} - \frac{1}{\rho} d^m_{:s} \right) - \frac{p - p^m}{S^m}$

    end for

    $p^m \leftarrow \sum_{s=1}^{S^m} \theta^m_{rs}$ ▷ Update the mth marginal

  end for

end while